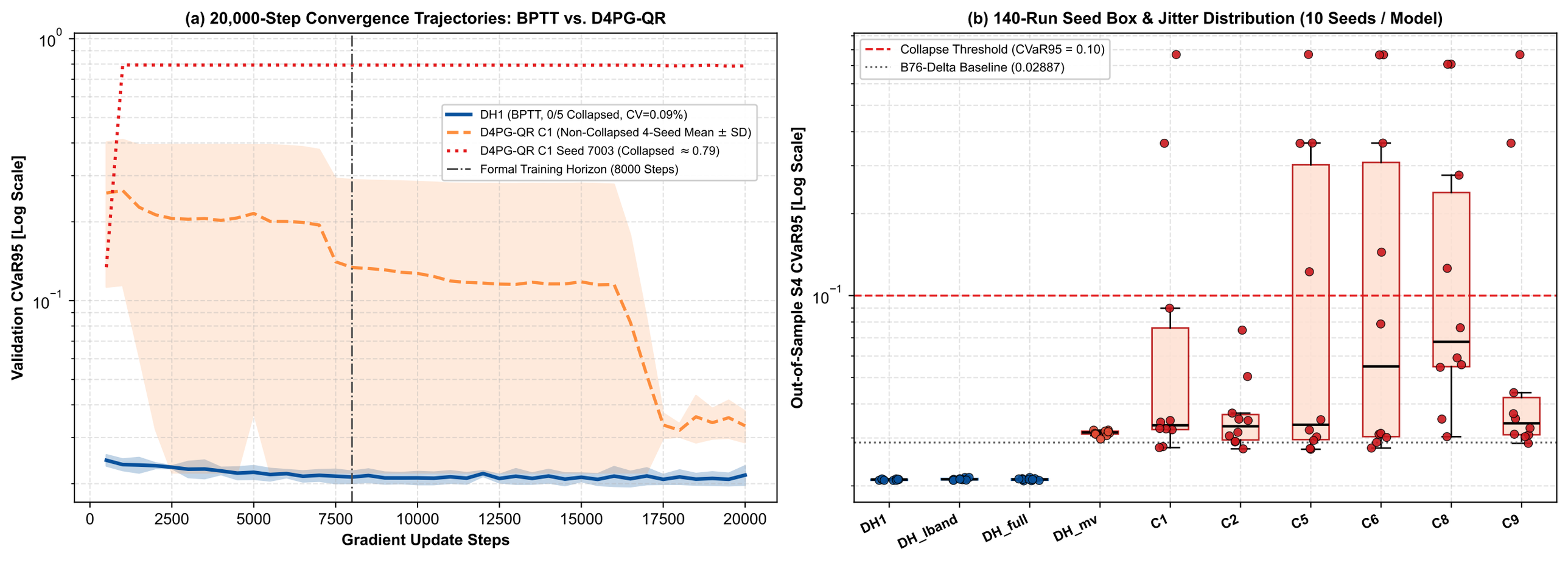
(a) 20,000-Step Convergence Trajectories: BPTT vs. D4PG-QR
Validation CVaR95 [Log Scale]
10^{0}
10^{-1}
DH1 (BPTT, 0/5 Collapsed, CV=0.09%)
D4PG-QR C1 (Non-Collapsed 4-Seed Mean ± SD)
D4PG-QR C1 Seed 7003 (Collapsed ≈ 0.79)
Formal Training Horizon (8000 Steps)
0
2500
5000
7500
10000
12500
15000
17500
20000
Gradient Update Steps
(b) 140-Run Seed Box & Jitter Distribution (10 Seeds / Model)
Out-of-Sample S4 CVaR95 [Log Scale]
10^{-1}
Collapse Threshold (CVaR95 = 0.10)
B76-Delta Baseline (0.02887)
DH1
DH_lband
DH_full
DH_mv
C1
C2
C5
C6
C8
C9